Homework

Homework 10 — Points of a tetrahedron — 50 min.

Consider a non-degenerate tetrahedron $ABCD$. The points I, J and K are the midpoints of $[AB]$, $[AC]$ and $[AD]$ respectively.

1. Express $\overrightarrow{IJ}$ in terms of $\overrightarrow{BC}$ and $\overrightarrow{IK}$ in terms of $\overrightarrow{BD}$.
2. Deduce that the planes (IJK) and (BCD) are parallel.
3. The point G is defined by $\overrightarrow{AG} = \frac{1}{4}(\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD})$. Does the point G belong to the plane (IJK)? Justify.
4. For this question only, assume that, in an orthonormal frame, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$ and $D(0;0;4)$. Calculate the lengths IJ, IK and JK.